Xianliang Huang

World Model & 3D Vision Researcher | Ph.D., Fudan University
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Research interests: Physical intelligence, world models, 3D reconstruction, neural rendering, Gaussian Splatting, NeRF, physical inverse problems

Research Profile

I build physics-grounded 3D world models for reconstructing, restoring, and predicting dynamic physical scenes from multi-view visual data. My background connects mathematics, wireless physics and electromagnetic inversion, computer graphics, neural rendering, 3D scene reconstruction, and industrial-scale world model engineering at ByteDance.

Education

- Fudan University, School of Computer Science and Technology - Ph.D. in Electronic Information / Computer Science, 2021.09-2025.06. Advisor: Prof. Shuigeng Zhou. Direction: multi-view image generation, restoration, and reconstruction.
- Xiamen University, School of Electronic Science and Technology - M.S. in Wireless Physics, 2018.09-2021.06. Advisor: Prof. Qinghuo Liu. Direction: electromagnetic inversion and AI-assisted physical modeling.
- Nanchang University, School of Mathematics and Computer Science - B.S. in Mathematics and Applied Mathematics, 2014.09-2018.06.

Professional Experience

- ByteDance - World Model Algorithm Researcher, 2025.07-Present. Large-scale fisheye scene reconstruction; feed-forward 3DGS cross-device mapping and localization; long-sequence offline and incremental reconstruction; 3DGS/4DGS scene generation and reconstruction.
- CASCO Signal Ltd. - Large Model Algorithm Research Intern, 2025.03-2025.06. AI algorithms and technical reports for rail-transit scenarios; RAG optimization for diagnostic large models.
- Hikvision - Algorithm Intern, 2024.03-2024.09. Open dataset parsing; EVA-CLIP deployment and API documentation; detector training and fine-tuning; data augmentation, data quality improvement, and label semantic graph construction.
- Beijing Shenshang Technology - Algorithm Researcher, 2021.06-2021.09. 3D digital human technology for virtual anchor generation; GAN and classical face reconstruction methods for 3D face reconstruction and multi-view face synthesis.

Selected Publications and Accepted Papers

- Xianliang Huang, Jiawen Li, Yanjin Chen, Feng Han, Qinghuo Liu. Hybrid Electromagnetic Inversion of Irregular Scatterers Embedded in Layered Media by VBIM and MET. IEEE Transactions on Antennas and Propagation, 68(12):8238-8243, 2020.
- Xianliang Huang, Jiawen Li, Jianliang Zhuo, Feng Han, Qinghuo Liu. Fast and Reliable Reconstruction of 3-D Anisotropic Objects Buried in Layered Media by Cascaded Inverse Solvers. IEEE Geoscience and Remote Sensing Letters, 2021.
- Xianliang Huang, Zhizhou Zhong, Shuhang Chen, Yi Xu, Jihong Guan, Shuigeng Zhou. NeRF-MIR: Neural Radiance Fields for Masked Image Restoration. IEEE Transactions on Neural Networks and Learning Systems, 2026.
- Xianliang Huang, Jiajie Gou, Shuhang Chen, Zhizhou Zhong, Jihong Guan, Shuigeng Zhou. IDDR-NGP: Incorporating Detectors for Distractor Removal with Instant Neural Radiance Field. ACM Multimedia 2023. Oral.
- Shuhang Chen*, Xianliang Huang*, Zhizhou Zhong, Jihong Guan, Shuigeng Zhou. Accurate Anthropometric Measurement Extraction with Focused Human Body Model. CVPR 2025.
- Xianliang Huang, Yining Lang, Ying Guo, Yuan He, Hui Xue, Li Zhao, Shuigeng Zhou. DR-Net: A Multi-View Face Synthesis Network Driven by Dual Representation. IEEE ICME 2023. Oral.
- Xianliang Huang, Chen Xiao, Yuanxiang Ni, Guanming Liu, Mingkai Liu, Dikai Fan, Xiao Liu, Hao Zhang. Semantic-Guided Progressive Object Removal with Gaussian Splatting. ICRA 2026. Accepted.
- Mingkai Liu, Dikai Fan, Haohua Que, Haojia Gao, Xiao Liu, Shuxue Peng, Meixia Lin, Shengyu Gu, Ruicong Ye, Wanli Qiu, Handong Yao, Ruopeng Zhang, Xianliang Huang. MACE: Mixture-of-Experts Accelerated Coordinate Encoding for Large-Scale Scene Localization and Rendering. ICRA 2026. Accepted.

- Xianliang Huang, Shuhang Chen, Zhizhou Zhong, Jiajie Gou, Jihong Guan, Shuigeng Zhou. Hi-NeRF: Hybridizing 2D Inpainting with Neural Radiance Fields for 3D Object Removal. ACCV 2024.

Manuscripts under Review

- Xianliang Huang, Chen Xiao, Shuhang Chen, Jiajie Gou, Jihong Guan, Shuigeng Zhou. A Cascade 3D Inpainting Scheme Combined Multi-view Diffusion Prior with Gaussian Splatting. Under review at IEEE Transactions on Cybernetics.
- Xianliang Huang, Chen Xiao, Mingkai Liu, Jingke Zhou, Dikai Fan, Yuewen Ma, Xiao Liu, Shuigeng Zhou. FiSh3R: Visual Geometry Transformer for Distortion-Aware Fisheye Reconstruction. Submitted to CVPR 2026.
- Jingke Zhou, Chenhang Ma, Zhizhou Zhong, Mingkai Liu, Zhuang Zhou, Yichan Ji, Yaoyuan Zhang, Dikai Fan, Binghua Su, Bo Cai, Xianliang Huang*. Think Locally, Refine Globally for Memory-Efficient Streaming 3D Reconstruction. Submitted to ICML 2026.
- Qize Jiang, Xianliang Huang, Hao Zhang, Weiwei Sun. Separable Policy Learning for Emergency Vehicle Prioritized Traffic Signal Control. Submitted to ICML 2026.
- Xianliang Huang, Jiajie Gou, Mingkai Liu, Qize Jiang, Dikai Fan, Yuewen Ma, Xiao Liu, Hao Zhang. SSOR-GS: Seamless Small Objects Removal via 3D Gaussian Splatting. Submitted to ICME 2026.
- Xianliang Huang*, Chen Xiao*, Jiajie Gou, Yixin Ren, Jihong Guan, Shuigeng Zhou. LLMs Powers Controllable Distractor Generation and Removal with Gaussian Splatting. Submitted to ICASSP 2026.
- Jingke Zhou, Chenhang Ma, Mingkai Liu, Ruihan Hou, Weichen Zhan, Xianliang Huang*. Gaussian Eraser: Seamless 3D Scene Inpainting via Region Score Distillation. Submitted to ICME 2026.
- Yuanxiang Ni, Xianliang Huang, Chenhang Ma, Chen Xiao, Yuewen Ma, Ruxin Wang, Hao Zhang. CoGeo-GS: Concept-Driven and Geometry-Aware Multi-Object Removal in 3D Scenes. Submitted to ICME 2026.
- Mingkai Liu, Haohua Que, Dikai Fan, Xianliang Huang, Haojia Gao, Handong Yao, Shuxue Peng, Meixia Lin, Feng Shuo, Fei Qiao. ACEsplat: Fast Per-Scene 3D Gaussian Reconstruction from RGB Images and Poses. Submitted to ICME 2026.
- Guanming Liu, Meng Wu, Peng Zhang, Yu Zhang, Yubo Shu, Xianliang Huang, Ning Gu, Liuxin Zhang, QianYing Wang, Tun Lu. A Proxy Agent for Context-Aware Intent Understanding via Retrieval-conditioned Inference. Under review at ACM Transactions on Information Systems.

Research Projects

- National Key R&D Program: fast computation for multi-parameter scattering of airborne composite-field-source electromagnetic signals.
- National Key R&D Program: joint electromagnetic-field inversion methods and imaging techniques.
- Hikvision collaboration: scalar-vector collaboration for large-scale high-potential data.
- Human body measurement project: anthropometric measurement extraction from reconstructed human meshes with a large human body model.

Honors and Awards

- Outstanding Graduate, Fudan University; Outstanding Student, Fudan University; First-Class Doctoral Scholarship, Fudan University.
- Outstanding Student, Xiamen University; Outstanding Student Leader, School of Electronic Science and Technology, Xiamen University.
- Provincial First Prize and Provincial Second Prize, National Undergraduate Mathematics Competition.
- Second Prize, APMCM Asia-Pacific Mathematical Modeling Contest.
- Two Second Prizes, Fujian Graduate Artificial Intelligence Competition, 2019: license plate recognition track and multi-channel speaker verification track.

Core Technical Areas

Neural rendering; NeRF; 3D Gaussian Splatting; 4D Gaussian representation; multi-view generation, restoration and reconstruction; visual geometry models; diffusion priors; large-scale fisheye reconstruction; scene localization and rendering; physical inverse problems; electromagnetic inversion; RAG optimization for domain large models.